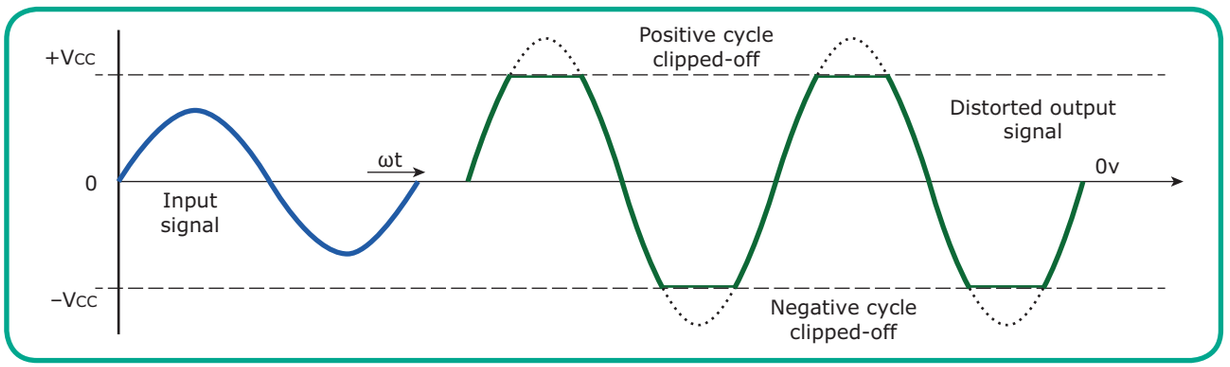


Figure 5.33 **Amplitude Distortion**

5.14.1 Amplitude Distortion

The amplitude distortion occurs, when the characteristic change of the amplifier is non-linear. This is shown in Figure 5.33

Figure 5.33 shows that the negative peak is flattened. This flattened portion is called as amplitude distortion. Low power supply and improper bias are the other reasons for this type of distortion. Since power amplifiers handles large signals and therefore, the problem of distortion immediately arises. For the comparison of two power amplifiers, the one which has the less distortion, is the better than the other.

5.14.2 Phase Distortion (or) Delay Distortion

Phase distortion is a type of amplifier distortion which occurs in a non-linear transistor amplifier when there is a time delay between the input signal and its appearance at the output.

5.14.3 Frequency Distortion

Frequency distortion is another type of amplifier distortion which occurs in a transistor amplifier when the level of amplification varies with frequency.

Noise: A noise is nothing but an unwanted voltage enters in the input of the amplifier. This may happen due to internal and external disturbances. Outside noises can be created through electrical devices such as motors, fluorescent bulbs. Internal noise can occur due to the inductor or coil used in the amplifier and this is termed as random noise.

The signal to noise ratio is one of the important factors to decide the quality of the amplifier. In order to eliminate the distortion and to increase the ratio between the signal and noise, negative feedback method is the best option.

5.15 APPLICATIONS OF A TRANSISTOR

1. When a transistor used for amplification purpose, it is called an Amplifier.
2. When a transistor acts as a switch, it is called a switch.
3. Transistor in active mode, (Analog Electronics).
4. Transistors in cut off saturation mode, (Digital Electronics).
5. Transistor can be used for an oscillation purpose, it is called as an Oscillator.
6. Transistors can be used from transistor radio to computer in all electronic circuits.

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